7. Capacitors

Follow charge through time, then prove where it went

A measured journey between empty and charged

Lesson 7 · three measured runs · isolated 3 V; adult controls energizing

What you need

- Switched holder with two AA cells
- 1000 μF electrolytic capacitor, 10 V or higher
- 10 kΩ and 22 kΩ resistors
- Digital multimeter with insulated leads
- Stopwatch, graph paper, removable +/− labels

Predict first

Imagine equal snapshots taken while an empty capacitor charges through a resistor. Will its voltage gain be the same in every snapshot, or will the gains shrink? Draw the rise. Beside it, draw what you expect after the battery is removed and the capacitor is allowed to discharge through the same resistor.

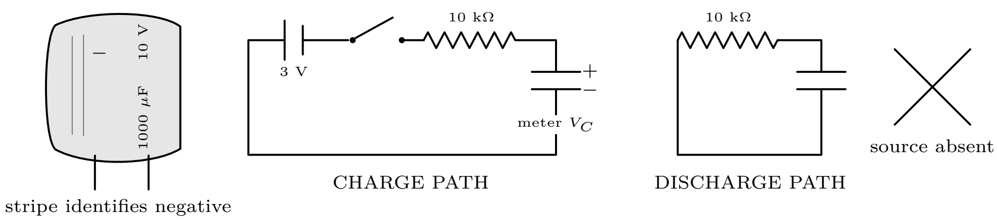
Four states, one permitted transition at a time

A COLD: source out; $V_C < 0.10$ V $\longrightarrow$ **B CHECKED**: polarity; rating; S open $\longrightarrow$ **C TIMING**: source in; hands away $\longrightarrow$ **D DISCHARGE**: source out; R across C $\longrightarrow$ A

THE CIRCUIT HAS ONLY ONE DISCHARGE TOOL

Never bridge capacitor leads with a screwdriver, wire, or probe tip. Remove the source, then leave a resistor across the capacitor until the meter reads below 0.10 V. A small part can still heat, leak, or rupture if reversed. Stop for warmth, swelling, odor, liquid, damage, or an unexpected reading.

Identify the real part before trusting the symbol



INSPECTION BEFORE RUN 1

☐ capacitor marked at least 10 V ☐ stripe goes to source negative ☐ red probe at + ☐ meter set to DC V ☐ switch open ☐ source not yet installed

Run 0 — learn the endpoints

1. With the capacitor absent, measure the source: $V_S =$ ________ V. Remove the holder.
2. Assemble the charge path. Confirm $V_C < 0.10$ V, then the adult installs the holder. Ask: *what sign should appear, and what would a minus sign tell us?*
3. Close the switch for 60 s. Record $V_{\text{high}} =$ ________ V. Open the switch; remove the holder; use the discharge path.
4. Do not change the circuit until $V_C =$ ________ V, below 0.10 V.

7. Capacitors

Repeated timing turns a curve into evidence

A measured journey between empty and charged

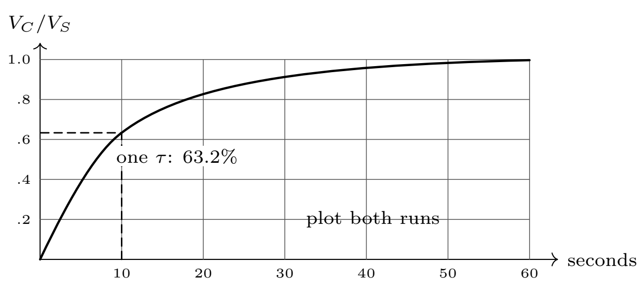
Run 1 — charge, twice

One learner calls the times, one reads the display, and one records. Keep probe clips fixed. At the call of “start,” the adult closes the switch. Record the first run, return to State A, then repeat without changing any part.

Time (s)	0	2	5	10	15	20	30	40	50
Run 1 V_C	____	____	____	____	____	____	____	____	____
Run 2 V_C	____	____	____	____	____	____	____	____	____
Mean V_C/V_S	____	____	____	____	____	____	____	____	____

Pause at 5 s, 10 s, and 20 s

Which interval added the most voltage? ____________ Did the second run reproduce the first? ____________ If V_C is closer to V_S, is the voltage remaining across the resistor larger or smaller? ____________



Calculate the landmark before comparing it with the graph

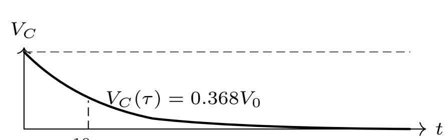
$$C = 1000\,\mu\text{F} = 0.001\,\text{F}, \qquad \tau = RC = (10{,}000\,\Omega)(0.001\,\text{F}) = 10\,\text{s}.$$

For charge, $V_C = V_S(1 - e^{-t/RC})$. Thus at 10 s the prediction is $0.632V_S$. With measured $V_S =$ ______ V, the predicted 10 s reading is _______ V. The two measured readings were ______ and ______ V.

Run 2 — remove the source and follow the energy

Charge for 50 s. Open the switch, remove the holder, and close only the resistor discharge path. Keep capacitor polarity and meter leads unchanged.

Time (s)	0	5	10	15	20	30	40	50
V_C (V)	____	____	____	____	____	____	____	____



7. Capacitors

Change one thing; keep every safety state

A measured journey between empty and charged

Run 3 — test whether RC really sets the clock

Substitute 22 kΩ for 10 kΩ; change nothing else. Perform two charge trials. Use fractional voltage rather than a guessed visual endpoint.

Run	R	C	calculated τ	$0.632V_S$	measured time to target	difference
1	10 kΩ	1000 μF	10 s			
2	10 kΩ	1000 μF	10 s			
3	22 kΩ	1000 μF	______			
4	22 kΩ	1000 μF	______			

Predict first

$22/10 = 2.2$. If $\tau = RC$, the target time should change by a factor of ________. Measured factor: ________. What stayed controlled? ____________________________________

Stored energy: calculate it, do not dramatize it

$$E = \frac{1}{2}CV^2 = \frac{1}{2}(0.001\,\mathrm{F})(3.00\,\mathrm{V})^2 = 0.0045\,\mathrm{J}.$$

Doubling voltage would multiply the energy by $2^2 = 4$, even though capacitance did not change. This experiment deliberately stays at 3 V and never produces a spark. The useful observation is the measured voltage history, not spectacle.

Questions that connect the curves to the circuit

1. Why is the charge curve steepest at the beginning? ____________________________________
2. Why does opening the source path not make V_C instantly zero? ____________________________________
3. During discharge, did the voltage sign reverse or keep its sign while its magnitude fell? ____________________
4. What evidence distinguishes "the capacitor is empty" from "the source is disconnected"? ________________________________

Diagnose only in State A

What the meter shows	De-energized check	Correction before another run
Negative sign	Probe placement and capacitor stripe	Red to labeled +; preserve capacitor polarity.
Near 0 V throughout charge	Switch path, clip seating, resistor bands	Restore the complete series path; never bypass R.
Starts above 0.10 V	Residual charge	Source out; resistor across C; wait and verify.
Curve too fast or slow	Actual R and printed C	Recalculate RC from the parts actually fitted.
Trials disagree sharply	Timing call and loose leads	Secure with switch open; restart both trials.
Warmth, swelling, odor, leak	No further test	Adult removes the holder and retires the capacitor.

POLARITY AND DISCHARGE CHECK BEFORE EVERY CHANGE

Open the switch. Remove the battery holder. Put the resistor across the capacitor. Watch V_C fall below 0.10 V. Read the meter sign and magnitude; never infer zero charge merely because the battery is absent.

Final proof

1. **Repeatability:** the two 10 kΩ values at 10 s differed by ________ V.
2. **Landmark:** mean $V_C(10\,\mathrm{s})/V_S =$ ________, compared with 0.632.
3. **Direction:** charge approached ________; discharge approached ________.
4. **Control:** changing only R multiplied the observed time by ________, so the evidence ☐ supports ☐ challenges $\tau = RC$.
5. **Safe finish:** source removed; resistor fitted; $V_C =$ ________ V < 0.10 V. Adult initials: ________.